Assignment 4: Argumentative Essay

The Value of Business Graduates in an AI-Enabled Economy

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Company A: Shell

Company B: KazMunayGas (KMG)

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1. Position Statement:

The Enhanced Premium on Human Agency Artificial intelligence (AI) is rapidly transforming modern enterprise operations by automating routine data processing, financial modeling, and algorithmic forecasting. However, rather than diminishing the worth of business professionals, AI will make business graduates significantly more valuable, not less valuable. In an AI-enabled economy, raw analytical speed and computational capacity are commoditized. The true bottleneck in organizational value creation shifts from quantitative execution to strategic contextualization, moral agency, legal accountability, and cross-functional governance. While AI models can synthesize financial statements and project operational scenarios, they lack legal liability, domain-specific contextual awareness, and socio-political empathy. Business graduates who possess durable human, digital, and meta-capabilities combining analytical fluency with institutional governance act as essential decisionmakers who validate, contextualize, and direct AI outputs toward sustainable economic value.

2. Financial Evidence & Discrepancy Verification:

Human Judgment in Capital Allocation The indispensable value of business graduates is demonstrated by quantitative analysis and discrepancy audits of financial data. In 2025, JSC NC KazMunayGas (KMG) reported consolidated revenue of 9,371 billion KZT ($17,977 million), EBITDA of 2,393 billion KZT ($4,591 million), and net profit of 1,072 billion KZT ($1,996 million) [1]. KMG maintained an exceptionally conservative Net Debt-to-EBITDA leverage ratio of 0.16x (with net debt at 375 billion KZT) [1]. During independent ratio analysis and AI discrepancy testing, generic AI models proposed unhedged retail price floating and mass divestment of mature upstream assets to maximize short-term free cash flow per share, imitating Shell plc's capital strategy (which generated $26.1 billion in free cash flow) [2]. However, a human analyst's discrepancy audit revealed that applying unconstrained AI optimization algorithms to KMG would ignore the company's statutory obligations to supply crude to domestic refineries at subsidized prices under Ministry of Energy Order No. 171 [3]. Furthermore, human verification of operational trials such as the Slim Hole drilling initiative at Uzen field which yielded a 30% CAPEX reduction (4.6 billion KZT savings) [1] highlights that human financial analysts are required to evaluate operational trade-offs, tax code constraints, and capital structure safety before committing capital expenditure.

3. Organizational & Workforce Evidence:

Governing AI in the Operational Model In organizational management, AI functions strictly as a Decision Support System (DSS), whereas human graduates retain exclusive accountability across the POLC framework. Within KMG's 52,000-worker operating footprint, workforce restructuring in the Mangystau region is categorized as Human Only because employment decisions directly impact regional socio-economic stability and require strict human management [4]. Operational predictive maintenance workflows such as the ABAI platform covering 1,660 waterflooding injection cells and predictive failure monitoring at the Atyrau Refinery (ANPZ) targeting a 15% reduction in unplanned downtime (baseline: 120 hrs/yr) [1], [5] operate under an AI Executes + Human Review allocation model. Here, a certified Chief Mechanical Engineer must physically inspect vibrometer telemetry and sign off before executing any SAP ERP work order. The computer vision TUMAR safety system across 60 workover crews similarly requires human HSE inspector verification [1]. As mapped in career capability portfolios, business graduates possess the meta-capabilities required to design governance risk registers, enforce manual fallback protocols during system outages, and prevent algorithmically induced operational shutdowns.

4. Comparative & Transfer Reasoning:

Contextual Boundaries of Strategy Comparative strategic transfer analysis between global energy major Shell plc and national champion KazMunayGas proves that business strategies cannot be blindly copied or algorithmically generated without contextual filtering [2]. Shell operates as an international capital allocator prioritizing shareholder distributions ($22.4 billion in 2025) and portfolio high-grading [2]. In contrast, Kazakhstan's sovereign energy anchor, fulfilling a mandatory dual mandate: generating commercial returns while guaranteeing domestic fuel security, maintaining 78% local procurement content, and financing regional infrastructure (such as the 121.2 billion KZT Kenderli desalination plant) [1]. An AI prompt recommending that KMG adopt Shell's strategy of floating domestic fuel prices and divesting lower-margin refining assets was explicitly REJECTED. Direct transfer would conflict with RK Ministry of Energy Order No. 171, trigger domestic inflation, and cause regional instability [3]. Business graduates provide the essential transfer reasoning required to adopt technical innovations (predictive maintenance, OGMP 2.0 methane tracking) while adapting climate frameworks (Zhanaozen 247 MW hybrid plant) and rejecting unsuited commercial practices.

5. Rebuttal of Counterargument:

The Illusory Automation of Judgment Critique of this position rests on the counterargument that advancing LLMs and autonomous AI agents will soon outperform human business graduates in financial analysis, market research, and operational planning. Proponents of this view argue that as AI automates complex analytical tasks at near-zero marginal cost, entry-level business roles will become obsolete, reducing overall graduate demand. While this counterargument accurately identifies the obsolescence of purely mechanical data processing, it fundamentally misinterprets the nature of business decision-making. AI models optimize within closed historical data distributions but struggle with novel geopolitical disruptions, regulatory shifts, and moral hazard. More critically, under corporate law and governance codes, an AI algorithm cannot hold fiduciary duty, sign statutory compliance documents, or assume legal liability for operational failures. Automating a non-bottleneck station accelerates Work-In-Progress (WIP) accumulation

without increasing process throughput. When AI generates voluminous analytical outputs, the operational bottleneck shifts directly to human verification, risk auditing, and executive decision-making. Consequently, AI amplification elevates the business graduate from a routine data processor to an essential governance anchor.

6. Conclusion & Boundary Condition

In conclusion, AI technology does not replace business graduates; it shifts their value proposition toward higher-order governance, ethical stewardship, and contextual strategy. Business graduates who master human-AI collaboration will command a premium in the labor market by bridging raw computational power with institutional accountability. My judgment would change only under the explicit condition that autonomous AI systems were granted full legal personhood, moral agency, and independent fiduciary liability under Kazakhstan and international law, capable of autonomously assuming liability for operational or environmental outcomes without human oversight. Until such legal and ethical frameworks exist, business graduates remain the irreplaceable core of enterprise governance.

References (IEEE Format)

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